A HOUSE DIVIDED

Build Plan, Start to Finish

How to build an original American Civil War grand-strategy and real-time-battle game from the two research volumes: agent-built, clean-room, usage-efficient, gate-controlled.

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1. The plan on one page

What gets built. The game specified in Part XIV of the Civil War volume: a weekly-turn strategic campaign (supply, rivers and rail, politics, intelligence, named commanders) joined to real-time battles with thousands of individually simulated soldiers, smoke, command friction and morale, resolved by one deterministic simulation that also powers quick-resolve and AI forecasting.

How it gets built. Simulation first, tests first, data first. The deterministic core and its headless harness exist, and are proven repeatable, before any art, engine or UI is committed to. Every later layer is a thin shell over that core, so the riskiest 20 percent (scale, determinism, performance) is retired in the first 8 weeks.

Your three requirements and the mechanism that delivers each

Requirement	Mechanism	Where
Efficient with your usage	Research extracted once into schema-validated data (the PDFs are never loaded again); one short session per ticket with only its spec section; cheap models for mechanical work; heavy checks run in CI, not in chat; per-ticket usage ledger with stop-loss; no polling.	Sections 6, 9
Ironclad QC	A seven-layer gate stack (local, PR, merge, nightly, cycle, human, release); acceptance tests written from the spec by a different agent than the implementer; determinism hashes on every merge; historical calibration against documented outcomes; clean-room scans on every merge; every escaped bug becomes a permanent test.	Sections 10, 11
Steady but rapid progress	Two-week cycles that each end in a runnable, measured build; main always playable; risk-first ordering; WIP limits; explicit go/no-go gates; a pre-agreed descope ladder so schedule pressure never cuts quality gates.	Sections 7, 12

Honest bounds (read these before approving)

- **Fidelity is a funding question, not just a coding one.** Part XII's benchmark for a Total-War-beating target is 40 to 80 people and 36 to 48 months. Agents compress code, tooling, data and test work sharply. They do not compress motion capture, hero art, music, voice, historian review or legal clearance. The plan runs those on a separate human/vendor lane (Section 4).
- **Three ship points, not one date.** S1 Playable Battle (placeholder art, about week 22), S2 Campaign Alpha (about week 60), S3 Beta (about week 100). These are planning estimates with low confidence beyond S1; the plan re-baselines them with you at every gate.
- **"Ironclad" means measured and bounded, not zero defects.** The plan makes each class of defect detectable by a named, automated check and states the residual risks (Section 10.7).
- **This plan was written by a session that read the Total War encyclopedia and its audit.** Under Part XV that places it in the reference zone. It contains no Total War constants, names or schemas, but no implementer session may be this one (Section 11).

Decisions needed from you (everything else has a default)

#	Decision	Default if you say nothing
D1	Weekly usage budget and profile (Lean / Standard / Sprint, Section 9)	Standard, re-tuned after 10 tickets
D2	Where the code lives	A new private repository (this repo is unrelated and its CI is red on an unrelated runner issue)
D3	Core language	Rust core + C ABI for the engine host
D4	Who owns a GPU workstation for renderer work and visual QC	You or a rented GPU runner, from week 4
D5	Budget for the human/vendor lane (counsel, mocap, art, audio, historians)	Phase-gated; nothing committed before G0
D6	Game name	Keep the placeholder; run a trademark search before any public use

2. The standards this plan builds to

The brief behind the Civil War volume (quoted in Part XIV sections 6, 7, 9 and 15) asks for a game that beats the Total War series on graphics, audio, animation, soldier modelling, combat physics and AI, with an original design and a research base that is honest about its evidence. Your three build requirements are added to it. Part XIV states that each of its goals must become a testable criterion "with a number set by the prototype, not a slogan". The table below does that: the thresholds are **starting hypotheses**, locked or revised at gate M1 (week 8) and again at S1.

ID	Standard	Acceptance test (hypothesis until M1)	Verified by
S1	Graphics: more distinct soldiers on screen than the reference series, stable frame rate, smoke that affects play	20,000 soldiers visible across three render tiers at 60 fps on the recommended-spec reference PC; smoke density field changes line of sight in the simulation; per-soldier appearance variation set with at least 6 independent axes	Render benchmark scenes in CI GPU runner; scripted camera flights
S2	Audio: distance delay, aggregated musketry, terrain acoustics, acoustic shadow as a game factor	Sound arrival delay equals range divided by the speed of sound in 100% of tested shots; acoustic-shadow scenarios reproduce "inaudible at N miles" cases; voice budget never exceeded in the heaviest scene	Unit tests on the acoustics model; audio capture diff tests
S3	Animation: context-aware, motion-matched near soldiers, terrain-adaptive	No foot-slide above threshold on slopes in automated test scenes; reaction coverage list (flinch, take cover, aid wounded, fall) complete; animation variety metric (distinct clip + phase offset combinations per regiment) above target	Automated pose checks; human review each cycle
S4	Soldier modelling and combat physics: every shot simulated, deterministic	Per-projectile model in the engagement radius (fidelity class F0); replay hashes identical across 3 operating-system/compiler builds for 10,000 recorded battles; calibrated within tolerance of documented casualty-by-range and rate-of-fire targets for at least 8 reference engagements	Golden-replay suite; calibration suite
S5	AI: no blobbing, chokepoint stalls, terrain blindness; sensible retreat and siege handling	Pass the 12-point genre failure checklist (abstract lessons, sanitised) in at least 90% of scripted scenarios; AI uses the same command-message interface as the player and never reads hidden state	Nightly 1,000-battle AI runs; scenario bots
S6	Quick-resolve: same rules as played battles	Quick-resolve and played battles produce identical results from identical inputs; at least 20x real time for a 10,000-soldier battle on the reference CPU; fidelity class fixed per scenario, never chosen by hardware	Equality tests; bench
S7	Strategy depth: logistics, politics, intelligence, command friction are first-class	A scripted playthrough that wins every battle but ignores supply, politics and manpower loses the war; each pillar has at least one decisive scenario test	Campaign bot suite
S8	Historical rigor	100% of numeric records carry source and confidence; any record tagged unverified is excluded from default scenarios; two-source rule enforced by CI	Data validator; sampled re-verification
S9	Original work, clean room	All gates G0-G7 passed; zero open contamination findings at any milestone	Scanners; counsel
S10	Respect for the subject (Part XIV section 14)	Advisory-board sign-off at P2, P4, P5; Black Americans have agency systems in every campaign; no gratuitous depiction	Human review checklist
B1	Efficient with your usage	Tracked tokens-per-accepted-ticket trend flat or falling after week 6; no session exceeds 2x its estimate without a re-plan	Usage ledger
B2	Ironclad QC	Main is always green; zero open P0 defects; every escaped defect has a regression test within the cycle	CI; bug ledger
B3	Steady, rapid progress	A runnable demo and metrics report every 2 weeks; no gate missed by more than 25% without a re-plan with you	Cycle reports

3. Scope: what exactly is being built, in tiers

Each tier is a complete, shippable-in-principle product. A tier is only started when the previous one passes its gate, so the project is never in a state where nothing works.

Tier	Contents	Ship point
T0 Prototype	Deterministic core v0, headless harness, replay and hashing, 20,000-agent crowd render test, smoke grid, acoustics prototype	M1 (week 8)
T1 Battle slice	One complete battle sector (for example the Cornfield sector at Antietam) with weapons, smoke, morale, command friction, terrain and cover, first animation pipeline, quick-resolve, tactical AI v1	S1 (week 22)
T2 Scenario pack	About 10 historical battles from the datasets, operational AI, scenario editor and loader, replays, balance by simulation	S1.5 (week 34)
T3 Campaign core	District map, supply and logistics, commanders and relationships, politics, intelligence and fog, campaign AI, battle hand-off, saves; a bounded campaign (for example Virginia 1862)	S2 (week 60)
T4 Grand campaign	Full 1860-65 map and orders of battle, riverine and naval, audio and music production, localisation, accessibility, modding tools, multiplayer battles (lockstep, optional)	S3 Beta (week 100)
T5 Release	Clearance (G7), launch, patch pipeline	Week 112+

Explicitly deferred or optional

- Console ports, campaign multiplayer, a language-model advisor and learned physics-based character control are R&D tracks behind interfaces. None is a ship dependency (Part XII risk 7, Part XIV section 7).
- Naval warfare enters at T4 and reuses the battle architecture with ships as larger agents (Part XIV section 4.10).
- The part of the research the audit found missing or unreliable (Section 6.3) gates the scenarios that depend on it, not the whole project.

Naming

"A House Divided" is a placeholder from Part XIV. The words "Total War" and variants must not appear in the name, store keywords, code, data or marketing (Part XV section 10). A trademark search is a G7 requirement, and a quick pre-check belongs in P0 so a rename never happens late.

4. Who builds what, and what the cloud container can and cannot do

Lane	Owns	Notes
Agents (cloud sessions)	Deterministic core, campaign logic, AI, data compiler and validator, scenario editor logic, tests and harnesses, CI, scanners, procedural audio and acoustics model, programmer-art assets and shaders, documentation, data extraction from the research volumes	Everything verifiable by automated tests. This is where the schedule compresses.
You	Decisions at gates, creative direction, playtesting, budget, final approval of every gate that says "human"	Time-boxed: about 20 minutes per week plus gate reviews (Section 12).
Human / vendor lane	Counsel and IP review, historians and advisory board, motion capture sessions, hero art, uniform and weapon reference scanning, voice performers, composer, platform certification	Long lead times; started early so they never block the critical path (Gantt, amber lane).

Container limits that shape the plan

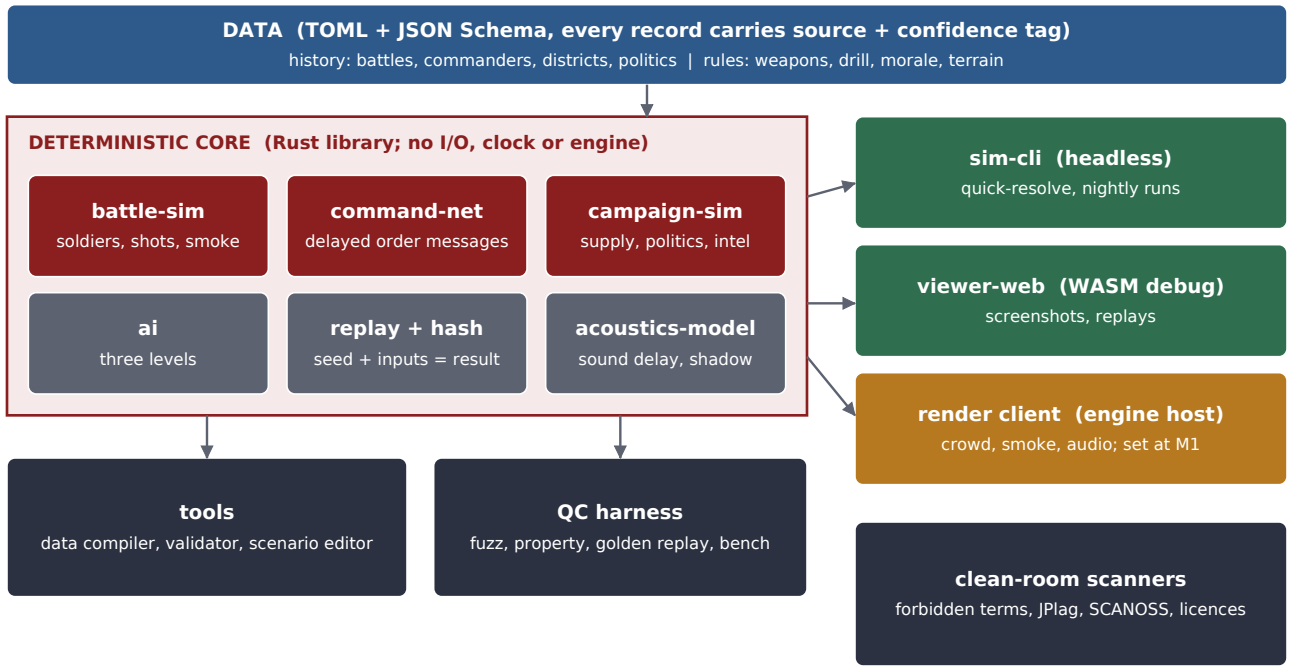
- The cloud container has no GPU and cannot host a large game engine. Core simulation, harness, tests and data work are fully verifiable there. A headless browser exists, so a WebAssembly debug viewer can produce screenshot-based visual regression checks for the simulation without a GPU.
- Renderer, shader and audio-mix work needs a GPU workstation or a GPU CI runner. That is why decision D4 and the amber box in the architecture diagram exist, and why the renderer prototype is on the gate M1 critical path.
- Outbound network from the container is restricted by policy, which is also what the clean-room rules want: implementer sessions should have an allowlist (package registries, engine documentation) and a denylist of the reference game and its mod sites.

Reconciling the volume's schedules

The volume gives two timelines that disagree. Part XIV proposes a prototype in months 0-2 and a vertical slice by month 6, with content ending at month 36. Part XII's planning scenario has pre-production of 9-12 months, a slice at 12-18, alpha near month 30 and ship at 36-48 for a 40-80 person team. XIV gives no team size. This plan resolves it as follows: the simulation milestone uses Part XIV's pace because agents can meet it; the renderer and content milestones keep a buffer drawn from Part XII because they depend on hardware, vendors and humans; and every date is a planning estimate re-baselined at each gate.

5. Architecture and technology

Deterministic core, thin shells: the core knows nothing about graphics, audio or the engine



Green = built and verified by agents in the cloud container. Amber = needs a GPU workstation (yours or a CI GPU runner).

5.1 Non-negotiable design rules

1. **One deterministic core.** A pure library with no I/O, no wall-clock, no threads that race, no unordered-map iteration in game logic. The same initial state, seed and input stream always give the same result.
2. **Fixed timestep and fixed-point authority.** Positions, velocities and ballistics use integer or fixed-point arithmetic in the authoritative state. Floating point is allowed only in non-authoritative visuals. This avoids cross-compiler drift and is verified by cross-platform hash tests from the first week.
3. **Seeded random streams per system** (ballistics, morale, AI) so adding a system never perturbs another's results.
4. **Snapshots and hashes every N ticks.** A replay is the initial state, the seed and the input stream. The nightly suite re-runs recorded battles and compares hashes.
5. **Orders are messages.** The player, the AI and (optionally) a language-model advisor all issue orders through the same delayed, lossy message interface. Nothing can issue a command that the rules would not allow a human.
6. **Data, not code, defines content.** Weapons, drill, formations, morale inputs, terrain, scenarios, commanders and districts live in schema-validated files that designers and mods can change.

5.2 Resolving the volume's simulation contradictions

The audit of the volume found three connected contradictions in Part XIV: it insists every shot is simulated while Part XII recommends a volley-level model for CPU cost; it says quick-resolve is "the same deterministic engine" while allowing a coarser tier when hardware cannot cope; and it never says who picks the tier. The plan resolves them with **fidelity classes**:

Class	What is simulated	Used for
F0 exact	Each projectile as a physical object (trajectory, wind, posture, cover, line of fire) for soldiers in or near engagement	Played battles inside the engagement radius; all calibration

Class	What is simulated	Used for
F1 bundled	Volleys as bundles along a firing line, hits applied statistically to the footprint under it, drawn from the same weapon tables and the same random streams	Distant mass fire in played battles; very large battles
F2 regiment	Regiment-level state advanced by rules derived from F0/F1 and checked against them	Strategic-scale quick-resolve of huge battles

- **The class is a property of the scenario and is recorded in the replay.** It is never chosen by the player's hardware, so multiplayer, replays and forecasts cannot diverge by machine.
- **Quick-resolve runs the scenario's own class headless**, so quick-resolve and played results match by construction. A calibration test keeps F1 and F2 within tolerance of F0 on shared scenes.
- The interface always shows which class ran.

5.3 Technology choices (recommendations with a decision gate)

Concern	Choice	Why
Core language	Rust workspace; C ABI (cbindgen) for engine hosts; WebAssembly build for the debug viewer	Compiler feedback catches defect classes early, which is cheaper than finding them in tests; strong tooling for property tests, fuzzing and mutation testing; WebAssembly lets agents inspect results without a GPU
Renderer host	Decide at gate M1. Default candidate from Part XII: Unreal Engine 5 with a custom crowd-render module; fallback: custom renderer over the same core	Part XII's matrix favours UE5 for lighting, tooling and hiring; the core is portable if the prototype disqualifies it
Data	TOML records plus JSON Schema; provenance block on every record	Human-readable diffs, easy validation, easy modding
Tests	cargo-nextest, proptest, cargo-fuzz, criterion benches, cargo-mutants, miri for unsafe boundaries	Standard Rust QC stack
CI	Pinned runner images and toolchains. Never `latest`.	The red check on the open PR is exactly this: an unpinned image lost a Python version
Supply chain	cargo-deny, an SBOM per build, JPlag (local) and SCANOSS, a custom forbidden-terms scanner	Part XV section 7; MOSS is excluded because it uploads code and is non-commercial
Telemetry	In-game overlay with per-system millisecond budgets; Tracy or equivalent traces	Part XII section 9.2

> **Why simulation first, renderer second.** The simulation is the part agents can build and prove alone, and it is the part that decides whether the whole concept is feasible (20,000 agents, deterministic, shared by play and quick-resolve). Rendering fidelity is the part that most needs people and hardware. Building them in that order means a failure of the concept shows up in week 8, not month 14.

6. From the two PDFs to a working data layer (extract once, never re-read)

The Civil War volume is 524 pages and roughly 1.7 million characters of text. Loading it into a session costs a large share of a usage budget every time. The plan therefore converts it **once** into small, schema-validated, provenance-tagged records and never loads the PDF into a build session again.

6.1 Mapping

Source (PDF part, pages)	Becomes	Used by	QC
Civil War vol. Parts II-VI (battles, pp. 28-257)	`history/battles/*.toml`: dates, forces, orders of battle, casualties as ranges with source and tag	Scenarios, calibration targets	Two-source rule; casualty sums checked by script
Parts VII-VIII (commanders, pp. 258-329)	`history/commanders/*.toml`: career, rank history, traits derived from the record, relationships	Command friction, AI personalities	Dates and ranks cross-checked; traits cite evidence
Part IX (politics, home front, pp. 330-375)	`history/politics/*.toml`: events timeline, cabinets, governors, finance series, factions	Campaign political layer	Timeline consistency tests (the audit found three misdated events)
Part X (gazetteer, pp. 376-411)	`history/districts/*.toml`: state, counties, rail, rivers, population and enslaved share	District map seed (400-600 districts)	Census figures validated against totals
Part XII (technology dossier, pp. 445-471)	Engineering specs and the M1 prototype brief	Renderer, audio, animation specs	Every number tagged; hypotheses stay hypotheses
Parts XIII, XV (legal, clean-room)	Process documents, gate definitions, scanner configuration	Section 11	Counsel review at G0
Part XIV (game concept, pp. 506-516)	Seed for the design specification (rewritten, not copied; Section 8 workflow)	Zone S specs	Spec editor + acceptance tests
Total War encyclopedia (78 pp.) and Part XI audit	**Restricted.** Only a one-page sanitised genre failure checklist of abstract lessons crosses the gate (no names, constants, schemas or formulas)	AI and QC test cases only	G2 sanitisation gate (scan + human)

6.2 Extraction pipeline (about one week of agent time, mostly cheap models)

1. Per-page text already exists. A script splits it by the volume's own section headings into one file per section.
2. Small-model agents convert each section into records that must validate against the JSON Schema; the validator rejects any record missing `source: {part, page}` or `confidence`.
3. The volume's tags map to `verified / partial / unverified`. **The audit showed that some "verified" tags sit on figures that are wrong**, so no record is promoted to `verified` by the extraction step: promotion needs a second independent source (a different agent reading a different source) recorded in the record.
4. Scenario loaders choose conservative values and exclude `unverified` records by default (Part XIV risk: historical data quality).
5. A sample of 5% of records per batch is re-verified by a second agent against sources; the error rate decides whether the whole batch is re-done.

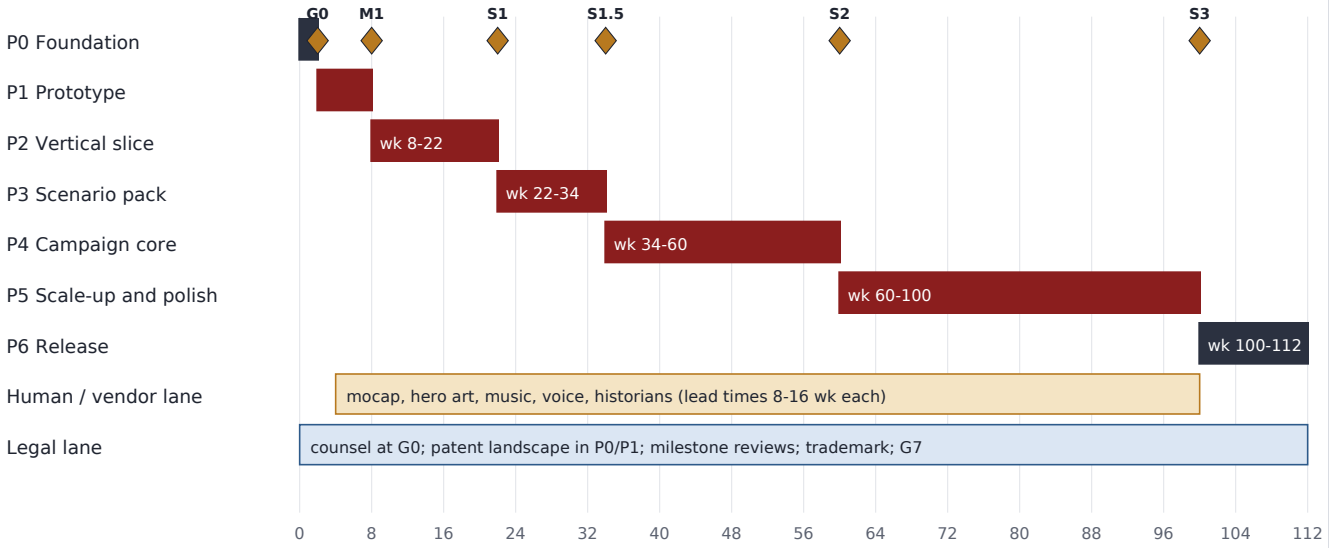
6.3 Known data problems carried in from the audits (do not seed until verified)

The audits found errors and conflicts that must not leak into the game's data. Each becomes a regression fixture: a test that fails if the unverified value is ever loaded as verified. Values below are the reviewers' best-knowledge corrections and **need two-source confirmation before use**.

Item	Volume says	Likely correct (verify)	Pages
Second corps commander killed at Antietam campaign	Reno "second", Mansfield after	Reno 14 Sep (first); Mansfield 17-18 Sep (second)	p.57
Pea Ridge dates	6-8, 8-9 and 7-8 March in different places	7-8 March 1862	pp.215, 270, 399
Mill Springs date	20 January (p.268) vs 19 January elsewhere	19 January 1862	pp.125, 268, 324
Davis captured	9 May (p.302) vs 10 May elsewhere	10 May 1865	pp.110, 302
Quantrill commissioned	after First Independence "March 1862"	First Independence was August 1862	pp.226, 323
Fort Fisher land face	480 yards "about 2,500 feet"; 47 guns land face	480 yd = 1,440 ft; 47 is about the total armament	p.111
Fort Fisher II Union force	8,000 (p.198) vs 9,632 (pp.111, 248)	Reconcile from the order of battle	pp.111, 198, 248
Hancock 1880 popular margin	about 40,000	under 10,000 (about 0.1%)	p.270
Wilson's Creek Union casualties	285 killed, total 1,317	258 killed makes the total sum	p.215
Gettysburg Confederate brigades (Stonewall, Florida, Georgia)	Brigades and commanders conflated	Rebuild from a published order of battle	pp.310-327
Part V Red River to Resaca	Sections missing (text splice)	Re-research Pleasant Hill to Resaca before using those scenarios	p.179
Island No. 10, Knoxville, Vicksburg and Port Hudson dates	Different spans in different places	Resolve to one cited span each	pp.122, 129, 152-154, 164

7. Roadmap: phases, exit gates and the critical path

Calendar view (weeks from start). Planning estimates, re-baselined at every gate.



Order is risk-first: determinism, scale and performance before content; content before polish; legal work in parallel from day one, not at month 18 as Part XV's work-breakdown table implies (Part XIII itself says counsel and a patent landscape come before building).

Phase (weeks)	Agent work	Human / vendor / legal lane	Exit gate (all must pass)
P0 Foundation (0-2)	Repo, pinned CI, schemas and validator, extraction pipeline pass 1, scanners, ticket and ledger tooling, forbidden-terms list (reference environment), project `CLAUDE.md` and ADR log	Engage counsel; draft independence charter (G0); commission patent landscape; pre-check name	**G0/M0:** CI green on an empty core with cross-platform determinism test wired; charter approved; ledger live
P1 Prototype (2-8)	Core v0: fixed-point math, RNG streams, agent store, spatial index, movement and formations, F0 ballistics from weapon tables, morale v0, command messages, replay and hash, headless harness; smoke grid and acoustics prototypes; 20,000-agent crowd render test on a GPU machine	Reference-weapon and drill sources gathered; mocap vendor shortlist; GPU workstation ready	**M1:** 20,000 agents tick within the CPU budget; replay hashes identical on 3 builds; render prototype measured on target hardware; engine decision recorded; S1-S9 thresholds locked
P2 Vertical slice (8-22)	One full battle sector: weapons, smoke, terrain and cover, entrenching, morale and routing, command friction, tactical AI v1, quick-resolve, animation pipeline with first captured clips, audio engine, calibration suite on at least 3 engagements, debug tools	First mocap session; first historian review; milestone legal review (G6)	**S1:** playable slice meets S1-S6 thresholds on reference hardware; calibration within tolerance; zero P0; G6 passed
P3 Scenario pack (22-34)	About 10 battles from data, operational AI, scenario editor and loader, replay viewer, balance by simulation, performance tiers	Advisory-board review of scenario framing	**S1.5:** every scenario plays end to end by bot and by you; nightly suite stable for 14 days
P4 Campaign core (34-60)	District map, supply and logistics, commanders and relationships, politics, intelligence, campaign AI, battle hand-off, saves	Historian and sensitivity review of political systems; music and voice contracts	**S2:** S7 tests pass; a bounded campaign is winnable and losable; G6 passed
P5 Scale-up and polish (60-100)	Grand campaign data, full orders of battle, riverine and naval, localisation, accessibility, modding tools, optional lockstep multiplayer, optimisation to the minimum-spec tier	Art and audio production at volume; beta programme	**S3 Beta:** all S and B criteria met; crash-free sessions at target; G6 passed

Phase (weeks)	Agent work	Human / vendor / legal lane	Exit gate (all must pass)
P6 Release (100-112)	Release builds, store compliance, patch pipeline	Trademark and patent clearance; AI-disclosure compliance (G7)	**G7:** counsel and producer clearance

Critical path

Core determinism (P1) → fidelity classes and calibration (P1-P2) → vertical slice (P2) → data-driven scenarios (P3) → campaign layer (P4). The renderer prototype runs in parallel with the core in P1 and rejoins at M1. The human/vendor lane starts at week 4 so mocap and art arrive when P2 needs them.

8. The agent workflow: one loop for every piece of work

1. **Spec section** (reference/spec environment): a short, numbered section with stated derivations, tunables and acceptance tests written as executable checks. The spec editor agent verifies it is testable and independent of restricted material.
2. **Ticket**: one session of work, one spec ID, fewer than about 400 changed lines, written from a fixed template (Appendix B).
3. **Tests first, by a different agent.** A test-writer agent turns the acceptance checks into failing tests. The implementer may not edit them. Separate authors prevent one agent's misreading from being baked into both code and tests.
4. **Implementation in a fresh session** that is given the ticket, its spec section and the repo map only. It never sees restricted material.
5. **Fast local checks** (Q0). Nothing is pushed red.
6. **Pull request and CI** (Q1-Q2). The pipeline is the single source of truth for pass or fail.
7. **Independent review** by a reviewer agent of a different model family where available, reading only the diff and the spec.
8. **Merge, ledger entry** (estimated vs actual usage, defects found), and an ADR line if a decision was made.

Roles and model tiers

Role	Does	Model tier
Orchestrator	Owns this plan, ticket queue, gates and the ledger; never carries content between environments; small context	Strong
Spec / design agents	Original specifications and acceptance checks; spec editor checks them	Strong
Core implementers	Simulation core, determinism, unsafe boundaries, performance work	Strong
General implementers	Campaign logic, AI behaviours, tools, UI logic	Mid
Mechanical agents	Data conversion, boilerplate, docs, fixtures, simple refactors	Small, escalate on failure
Test writers / QA bots	Tests from specs, scenario bots, balance batches	Mid
Reviewers	Independent diff-versus-spec review	Strong, different family where possible
Red team / auditors	Clean-room similarity and contamination probing (isolated)	Strong
Historians	Datasets with citations; second-source verification	Mid, sources only

Parallelism rules

- The core's critical path is serial: one core ticket at a time on the same module. Independent tickets (data batches, tools, tests, docs) run in parallel in separate worktrees.
- The number of concurrent sessions is set by the budget profile (Section 9), not by how many could run.
- Subagents are for wide read-only sweeps and independent batches. They are not for small edits, which cost more to brief than to do.

Environments (simplified from Part XV for one human and agent sessions)

Environment	May read	Writes	Rule
R reference	Restricted, permitted, preferred sources	Sanitised notes, historical datasets	Anything that read the reference-game material lives here, including the sessions that wrote the volume and this plan; never reused as implementers
S/I build	Specs, datasets, permitted sources only	Specs, code, data, assets	Fresh sessions; allowlisted network; no web search for the reference game or its mod ecosystem

Environment	May read	Writes	Rule
A audit	Everything, kept apart from S/I authors	Scan and red-team reports	Holds the forbidden-terms and reference sets; never writes product code
L legal	Everything needed to decide	Approvals	Humans only

9. Usage efficiency: how the build stays inside your budget

9.1 Rules

1. **Extract once.** Research becomes data (Section 6). No build session loads a PDF.
2. **Small sessions, one ticket each.** Context is the ticket, its spec section and a repo map of at most about 150 lines. Hand-offs are a note of at most about 40 lines, never a transcript.
3. **Plan once, record decisions.** Architecture decisions are written as short ADRs so they are not re-argued in later sessions.
4. **Cheapest competent model first.** Mechanical work starts on a small model and escalates only on failure; strong models are kept for the core, specs, review and red-teaming.
5. **Heavy checks run in CI or nightly, not in chat.** Fast checks run locally with quiet output so only failures enter the conversation (failures first, tail only).
6. **Incremental builds and caches** for compilation; test selection by changed crate in Q0.
7. **Fail fast, stop-loss.** A session that reaches twice its estimate stops and re-plans instead of continuing.
8. **No polling.** CI results and review events wake the session; nothing sits in a loop waiting.
9. **Write tests from specs, not from code.** It is cheaper to generate tests once from a precise spec than to debug vague behaviour repeatedly.
10. **Batch your decisions.** One short weekly review replaces many interrupts.

9.2 Budget profiles

Profile	Concurrent sessions	Typical week	When to use
Lean	1 (core), +1 mechanical	3-5 accepted tickets; nightly suite on; no new research	Tight budget; protects QC gates first
Standard	1 core + up to 2 parallel	8-12 accepted tickets; data batches; one review sweep	Default
Sprint	1 core + up to 4 parallel	15+ accepted tickets; short bursts before a gate	Only for the last 2 weeks before a gate

Planning allocation of a week's usage (revised after 10 tickets of real data): implementation 40%, review and QC 20%, data and research 15%, specs 10%, reserve 15%. The reserve is spent only on re-plans and escaped defects.

9.3 The usage ledger

Every accepted ticket appends one row: ticket, class (XS/S/M/L), estimated and actual usage in the unit your account shows, defects found in review, rework count. Week 3 replaces estimates with measured values. A weekly summary shows usage per accepted ticket; a rising trend triggers a process fix before any further feature work.

9.4 Your own time

- About 20 minutes per week: approve the next cycle's ticket queue and answer the decisions list.
- Gate reviews at M1, S1, S1.5, S2 and S3: a one-page report plus a demo capture, then a go / re-plan / stop decision.
- Everything else has a default so work never waits on you.

10. Quality control: the gate stack



10.1 What runs where

Layer	When	Checks	Pass criterion
Q0 local	Before every commit	Formatting, strict lints, unit tests for touched crates, forbidden-terms scan, "no process text" lint (blocks phrases such as first draft, my earlier, TODO in docs and shipped strings)	All pass in under a minute; nothing red is pushed
Q1 pull request	Every PR	Full unit and integration tests; data schema validation; licence scan; SCANOSS and JPlag against internal corpus; determinism smoke (3 replays hash-equal); benchmarks against baseline; coverage floor on rules crates	All pass; benchmark regression under 10%; rules-crate line coverage at or above 90%
Q2 merge to main	Every merge	Cross-platform matrix (3 operating systems, 2 compilers) with hash equality; property tests at 100,000 cases; 10 minutes of fuzzing on parsers and loaders	Hashes identical across the matrix; no new crash found
Q3 nightly	Unattended	1,000 headless AI-versus-AI battles asserting invariants (soldiers conserved, no NaN or stuck agents, replays reproduce, memory flat); calibration suite versus historical targets; worst-case performance scenes; 2-hour soak; miri or sanitiser runs on unsafe code	No invariant violation; calibration inside tolerance; performance inside budget
Q4 cycle / milestone	Every 2 weeks; every gate	Mutation testing on rules crates; independent review of the cycle's diff against specs; clean-room red-team pass; AI failure-mode checklist; bot playthroughs; 5% data re-verification	Mutation score at or above 80%; zero open red-team findings; checklist at threshold
Q5 human	Every cycle; at P2, P4, P5	Your playtest; accessibility checks; historian and advisory-board review; sensitivity checklist	Sign-off recorded in the gate report
Q6 release	Before any public release	G7: trademark and patent clearance, compliance, final scans, compatibility matrix	Counsel and producer clearance

10.2 Rules that make it ironclad in practice

1. **Main is always green and always playable.** A red main stops all feature work until fixed.
2. **Defect severity.** P0 (crash, desync, data corruption, contamination) blocks every merge. P1 is fixed within the cycle. P2 is scheduled.
3. **Every escaped defect gets a regression test and a one-line cause in the ADR log** before it is closed.
4. **Authors of tests and code differ.** Reviewers are a different agent, and where possible a different model family.
5. **No threshold is edited to turn a gate green.** Changing a threshold requires an ADR and your approval at a gate.
6. **Never skip, disable or quarantine a test to get green.** A flaky test is a defect in the test or the code and is fixed.
7. **Spec drift is a defect.** If code and spec disagree, the spec is amended first, then the tests, then the code.

10.3 Determinism QC (the project's core risk)

- Record every test battle as seed plus inputs; the expected hash every 100 ticks is checked in as a golden file.
- Hashes are compared across operating systems, compilers and optimisation levels on every merge to main.
- Static rules in the core's lints: no wall-clock, no thread-local randomness, no unordered iteration, no floating-point in authoritative state.
- A cross-platform failure is a P0 that stops merges until the cause is found.

10.4 Calibration QC (historical realism by measurement)

Following Part XV section 12, combat numbers are calibrated against documented evidence rather than copied: rates of fire from drill manuals and trials, casualty-by-range from well-documented engagements, march rates, and the outcomes of the reference engagements. The nightly suite reports the model's error for each target. Targets carry the confidence tag of their source; unverified targets are reported but do not gate.

10.5 Data QC

- The validator fails the build on any record missing source, confidence or a valid schema.
- Casualty and strength records are checked by script (components must sum to totals; dates within ranges).
- Cross-record tests: the same battle, person or place gives the same date and name everywhere it appears (the audit found many such splits).
- The regression fixtures from Section 6.3 stay in the suite permanently.

10.6 Genre failure checklist (sanitised, from the audit)

The audit's catalogue of weaknesses in the reference genre becomes a 12-point list of abstract tests, released through the sanitisation gate: blobbing, chokepoint stalls, terrain blindness, silly retreats, siege and fortification handling, statistical shortcuts that disagree with played results, hidden stats that mislead the player, line-of-sight confusion, unfair difficulty bonuses, formation reshaping stalls, order-latency misreads and unreadable morale feedback. It names no product and no constants.

10.7 Residual risks (stated, not hidden)

Residual risk	Why it remains	Containment
Historical facts the agents cannot independently confirm	No outside sources were reachable in the audit environment	Two-source rule; sampled re-verification; unverified data excluded by default
Visual and audio quality judgements	Not fully automatable	Human playtest every cycle; advisory review at P2, P4, P5
Model exposure to the reference game	Training data cannot be controlled (Part XV limit 2)	Input control, scans, independent-creation dossiers, counsel review
Patents on genre mechanics	Independent invention is no defence	Landscape search in P0/P1; counsel review before design lock
Unknown unknowns at scale	20,000-agent deterministic simulation is unproven for this team	Prototype first; fallback tiers; gate M1

11. Clean-room and legal controls (adapted from Parts XIII and XV)

> **Not legal advice.** Part XV says no court has endorsed an "AI clean room" and that an agent-based process can imitate the discipline of one without claiming legal immunity. Counsel must approve the charter at G0. The controls below make the independence argument stronger and the trail evidential; they do not guarantee an outcome.

11.1 Input tiers

Tier	Examples	Rule
Forbidden	Reference-game code, extracted data, models, textures, sounds, UI art; mod packs with publisher assets; verbatim wiki or forum text as content	Never enters any environment
Restricted	The 78-page encyclopedia, Part XI's audit, developer talks and community formula documentation of the reference game	Reference environment only; passes to specs only as sanitised notes
Permitted	Historical sources, public-domain text and images, engine and computer-science documentation, open-licence libraries under tracked licences	Any environment
Preferred	Historical record, calibrated simulation, the project's own prototypes and playtests	Default basis for decisions

11.2 Gates, with corrected timing

Gate	When	Approver	Change from Part XV
G0 Charter	P0, before anything is built	Counsel + you	Counsel engaged at the start; patent landscape commissioned here, not at month 18
G1 Input approval	Before a source enters an environment	Reviewer	None
G2 Sanitisation	Each note from reference to spec	Human + scan	The same gate must also cover historical datasets that the reference environment produces (Part XV left that unscreened)
G3 Spec release	Each spec to implementation	Spec editor + you	None
G4 Code and data	Every merge	Automated + reviewer	JPlag and SCANOSS locally; MOSS excluded (non-commercial, uploads code)
G5 Asset provenance	Every asset import	Art/audio lead	Manifest and human-authorship record required
G6 Milestone legal	M1, S1, S1.5, S2, S3	Counsel	Runs at every milestone, not only at release
G7 Release clearance	Before any public release	Counsel + producer	Trademark, patent, compliance, AI-disclosure rules at the time

11.3 Practical controls

- **Fresh implementers.** Implementation sessions start from clean prompts that contain only specs, datasets and permitted material. The sessions that read the encyclopedia, the audit or the Civil War volume's reference-game sections are never reused to implement.
- **Provenance manifests** for every code file, asset and dataset: author (agent or human), inputs by identifier, tools and versions, gates passed, content hash. Commits carry the ticket and spec IDs.
- **Independent-creation dossiers** per major system: problem, historical sources, alternatives, chosen design, calibration experiments, dates.
- **Original vocabulary and schemas.** Spec authors invent field and system names grounded in period terms; the UI starts from user needs and period documents, not from other games' layouts.

- **Assets.** Human authorship for key creative assets and records of it (US copyright requires it); licensed or indemnified generators only; consenting, fairly paid performers; public-domain sources checked item by item.
- **Contamination response.** Stop and quarantine, preserve evidence, assess with counsel, rebuild from the clean spec with fresh agents, fix the process, never edit logs.

11.4 Contradictions in the volume this plan resolves

- Part XV puts the agents that produced the volume in the reference environment, yet Part XIV (the design) was written by the same orchestrator and embeds audit conclusions. **Resolution:** Part XIV is treated as a draft seed, rewritten into original specs by fresh spec agents through the sanitisation gate.
- Part XV makes the volume "Phase 0" input although Part XI reproduces reference-game constants. **Resolution:** the volume's research parts (II-X) enter as data; Parts XI and the encyclopedia are Restricted and never reach build sessions.
- Part XIII lists MOSS as a CI gate while saying it is unsuitable. **Resolution:** JPlag and SCANOSS only.
- Part XIII retracts the claim that NEC "won because it documented a clean room" but a later paragraph repeats it. **Resolution:** the plan does not rely on that claim.

11.5 Questions for counsel (from Part XV section 14, plus two)

- Is the charter-and-logs approach likely to persuade for an agent-assisted process, given model training exposure?
- How should the encyclopedia and audit themselves be handled (never shared with the build environment, per the plan)?
- Patent exposure for genre mechanics; rules for AI-generated assets in target markets; performer and advisor contract terms; period imagery and symbols by region.
- Is a pre-emptive patent clearance needed for the command-message and fidelity-class designs? Does the title need a different name from the placeholder?

12. Steady but rapid progress: cadence, metrics and control

12.1 The two-week cycle

Day	Activity	Output
1	Plan: you approve the ticket queue (about 20 min); orchestrator sizes tickets and sets concurrency	Cycle plan
2-8	Build: tests first, implement, review, merge; main stays green; risk-first ordering within the cycle	Merged tickets
9	Integration freeze: Q4 sweep (mutation, independent review, red-team, bots, data re-verification)	Q4 report
10	Demo capture, metrics report, retro: one process fix chosen	Runnable build + one-page report

12.2 Metrics (reported every cycle)

Metric	Green	Amber action	Red action
Accepted tickets vs plan	within 15%	Resize next cycle	Re-plan with you
First-pass CI success rate	above 80%	Review test quality	Stop features; fix process
Determinism pass rate (matrix)	100%	n/a	P0: merges stop
Sim tick time vs budget	under 80% of budget	Profile before new features	Optimise only until back under
Escaped defects per cycle	0 P0, at most 2 P1	Add tests for the class	Root-cause review
Data provenance coverage	100%	Block new records	Block merges
Calibration error vs targets	inside tolerance	Investigate model	Do not add content on top
Usage per accepted ticket	flat or falling	Check ticket sizing	Move to Lean profile

12.3 The descope ladder (agreed now, so pressure never cuts quality)

If a gate looks like it will slip by more than 25%, features are cut in this order, never the gates themselves:

1. Optional multiplayer.
2. Naval and riverine battles (keep riverine supply on the campaign map).
3. Grand campaign reduced to several bounded campaigns.
4. Mocap breadth: hand-keyed animation for secondary actions.
5. Far-tier visual fidelity and some weather states.
6. Localisation to fewer languages at launch.

> **Never cut:** determinism, the QC gates, the clean-room gates, calibration against history, the data provenance rules, and the respectful handling of the subject (Part XIV section 14).

12.4 Risk register (top items)

Risk	Trigger	Mitigation
20,000-agent deterministic sim too slow	Tick above budget at M1	Fidelity classes; tiered update rates; data-oriented layout; fallback target counts
Cross-platform drift	Any hash mismatch	Fixed-point authority; strict lints; matrix on every merge
Render prototype misses target	M1 measurement	Extend renderer verdict to week 14; custom-renderer fallback over the same core
Research data errors enter the game	Sampled re-verification above 5% error	Two-source rule; exclude unverified; re-do batch

Risk	Trigger	Mitigation
Contamination finding	Any scan hit	Section 11.3 response; fresh agents
Usage overrun	Ticket above 2x estimate	Stop-loss; Lean profile; descope ladder
Vendor lead times slip	Mocap or art late	Placeholder-first pipeline; start vendor lane at week 4
Scope creep (grand campaign, ML features)	Unplanned feature requests	Tier gates; R&D tracks behind interfaces
Sensitivity failure	Advisory-board objection	Early framing decisions; reviews at P2, P4, P5; stop-ship authority

13. The first 30 days, ticket by ticket

Concrete enough to start the day you approve. Each line is one ticket-sized unit; sizes are relative (XS/S/M/L) until the ledger gives real numbers in week 3.

Week	Tickets	Exit check
1 Foundation	T-001 repo + pinned CI skeleton (S) · T-002 charter draft and input-tier policy for counsel (S) · T-003 project CLAUDE.md, ADR log, ticket template, usage ledger (XS) · T-004 JSON Schemas v0 for battles, commanders, districts, politics (M) · T-005 extraction scripts that split per-page text by section (S) · T-006 forbidden-terms scanner and no-process-text lint (S) · T-007 decisions D1-D6 recorded (XS)	CI green on an empty workspace; schemas validate fixtures; you have approved D1-D6
2 Core skeleton	T-008 fixed-point math and RNG streams (M) · T-009 agent store and spatial index (M) · T-010 state hashing + replay format (M) · T-011 headless runner (S) · T-012 cross-platform determinism test in CI (M) · T-013 extraction pass 1: 5 battles, 30 commanders by small-model agents + 5% re-verification (M)	Three replays hash-equal on three builds; first data records validate
3 Soldiers and shots	T-014 formations and movement (L) · T-015 weapon table schema and F0 ballistics (L) · T-016 calibration harness skeleton (M) · T-017 start crowd render prototype on GPU machine (M) · T-018 ledger review: replace estimates with measured usage (XS)	Soldiers march and fire in the headless harness; ledger has real numbers
4 Morale, orders, nightly	T-019 morale v0 (M) · T-020 command message bus with latency (M) · T-021 sim tick benchmarks and budget file (S) · T-022 nightly Q3 job v1 (M) · T-023 first demo capture via WebAssembly viewer (M) · T-024 first cycle report and re-baseline with you (XS)	20,000 agents ticking in the harness; nightly stable; first cycle report delivered

Immediately, in parallel (low usage, long lead time)

- Engage intellectual-property counsel and commission the patent landscape (human lane).
- Run a quick trademark pre-check on candidate names (human lane).
- Shortlist motion-capture and reenactor partners; book the first session for week 12-16 (human lane).
- Provision a GPU workstation or rented runner for week 3 onward (D4).

Appendix A. Repository layout

```
<repo>/
  CLAUDE.md     (map + rules, at most about 150 lines)
  docs/         specs/ (numbered sections), adr/, dossiers/, gates/, cycle-reports/
  crates/       core-math, battle-sim, command-net, campaign-sim, ai, acoustics, replay
  shells/       sim-cli, viewer-web, engine-host/ (decided at M1)
  data/         schemas/, history/ (battles, commanders, districts, politics), rules/, scenarios/
  tools/         data-compiler, validator, scenario-editor, scanners/
  qc/           golden-replays/, calibration/, fixtures/ (audit regression fixtures), bots/
  .github/workflows/   Q1, Q2, Q3 pipelines (pinned images and toolchains)
  ledger/       usage.csv, bugs.csv, provenance manifests
```

Appendix B. Ticket template and definition of done

Ticket template

Field	Content
ID, class	T-NNN; XS / S / M / L; estimated usage
Spec reference	Spec section ID (the only design context the implementer receives)
Acceptance tests	Names of the failing tests written by the test-writer agent (implementer may not edit)
Scope fence	Files and crates allowed to change; everything else is out of bounds
Non-goals	What this ticket must not do
Stop-loss	Replan if usage reaches 2x estimate or if the change exceeds about 400 lines

Definition of done (all required)

1. All acceptance tests pass; no test weakened or removed; no threshold changed.
2. Q0 and Q1 pass on the PR; Q2 passes on merge.
3. Independent review found no blocking issue (or each was fixed).
4. Determinism hash unchanged unless the spec explicitly changes behaviour, in which case the golden files change in the same PR with a reason.
5. Data records touched carry source and confidence; no unverified record became verified without a second source.
6. No Restricted material in code, data, strings or docs (scanners green); provenance manifest updated.
7. No process text in shipped artefacts; ledger row appended; ADR written if a decision was made.

Appendix C. Source map

Plan section	Draws on
1-3 summary, standards, scope	Civil War volume Part XIV sections 1, 6, 7, 9, 12, 15; Part XII section 10; audit findings on contradictions
4 who builds, schedules	Part XII sections 9.4 and 10 (team size, timeline, prototype milestones); Part XIV section 12
5 architecture, fidelity classes	Part XIV section 4 and 8; Part XII sections 6, 8.6-8.7, 9; audit findings 2 and 3 on Parts XIV/XII
6 data layer	Parts II-X (battles, commanders, politics, gazetteer); audit lists of errors; Part XV section 5 (derivations)
7 roadmap	Part XIV section 12; Part XV section 12; Part XII milestones M1-M2; audit finding on counsel timing (Part XIII pp.481, 501)
8-9 workflow, usage	Part XV sections 3, 6; general delivery practice
10 QC	Part XII section 9.2; Part XV sections 7-8, 12; the Total War audit's failure-mode list (sanitised); audit of the Civil War volume
11 clean room	Parts XIII and XV; audit findings on zone, MOSS, NEC and phase timing
12 control	Part XIV section 13; Part XII risk register

Limits of this document. It rests on the two PDFs and on two audits that checked the PDFs only against themselves and the reviewers' own knowledge; no outside source was consulted. Dates and thresholds are planning estimates and hypotheses, and every number in the standards table is a starting value for gate M1. The legal sections are process proposals, not legal advice.